

Lecture Notes on NAND Flash Memory Technology

Chapter 4: Read Operations and Sensing

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1 4.1 Read Operation Principles

• Basic Read Mechanism

Definition 1.1 (NAND Read Operation). Current-based sensing to determine stored charge state:

- Apply V_{read} to selected word line
- Apply V_{pass} to unselected word lines
- Sense current through bit line
- Compare with reference current

Read current equation:

$$I_{\text{read}} = \begin{cases} I_{\text{on}} & \text{if } V_{\text{th}} < V_{\text{read}} \text{ (conducting)} \\ I_{\text{off}} & \text{if } V_{\text{th}} > V_{\text{read}} \text{ (non-conducting)} \end{cases} \quad (1)$$

Current ratio for reliable sensing:

$$\frac{I_{\text{on}}}{I_{\text{off}}} > 100 \text{ (minimum for robust detection)} \quad (2)$$

• Read Voltage Selection

Optimal read voltage:

$$V_{\text{read}} = \frac{V_{\text{th0}} + V_{\text{th1}}}{2} - \text{margin} \quad (3)$$

where V_{th0} and V_{th1} are adjacent threshold distributions.

For multi-level cells:

$$V_{\text{read},i} = \frac{\mu_i + \mu_{i+1}}{2} - k_{\sigma} \sqrt{\sigma_i^2 + \sigma_{i+1}^2} \quad (4)$$

2 4.2 Sense Amplifier Design

• Current Sensing Principle

Theorem 2.1 (Bit Line Voltage Development). Voltage change during sensing period:

$$\Delta V_{\text{BL}} = \frac{I_{\text{cell}} \cdot t_{\text{sense}}}{C_{\text{BL},\text{total}}} \quad (5)$$

where:

$$C_{BL,total} = C_{junction} + C_{metal} + C_{parasitic} \quad (6)$$

$$= n \cdot C_{junction} + C_{metal/m} \cdot L_{BL} + C_{coupling} \quad (7)$$

• Sense Amplifier Architecture

Current-mode sense amplifier:

$$V_{out} = A_{SA} \cdot (I_{cell} - I_{ref}) \cdot R_{load} \quad (8)$$

Sensitivity requirement:

$$S_{SA} = \frac{dV_{out}}{dI_{in}} > \frac{V_{logic}}{I_{min} - I_{max,leakage}} \quad (9)$$

• Sensing Speed Optimization

Sensing time constraint:

$$t_{sense} < \frac{C_{BL} \cdot V_{margin}}{I_{cell,min}} \quad (10)$$

Speed-power trade-off:

$$\text{Energy}_{sense} = \frac{1}{2} C_{BL} V_{pre}^2 + I_{bias} \cdot V_{cc} \cdot t_{sense} \quad (11)$$

3 4.3 Reference Generation

• Reference Cell Design

Reference current generation:

$$I_{ref} = \frac{I_{on} + I_{off}}{2} \cdot \alpha_{tracking} \quad (12)$$

Tracking coefficient:

$$\alpha_{tracking} = \frac{dI_{ref}}{dI_{cell}} \approx 0.5 - 0.7 \quad (13)$$

• Process Variation Compensation

Reference voltage with compensation:

$$V_{ref}(T, V_{cc}) = V_{ref,nom} + TC \cdot \Delta T + SC \cdot \Delta V_{cc} \quad (14)$$

Temperature coefficient:

$$TC = \frac{1}{I_{ref}} \frac{dI_{ref}}{dT} \approx -1.5 \text{ to } -2.0 \text{ mV}/^\circ\text{C} \quad (15)$$

4 4.4 Read Disturb Mechanisms

• Read Stress Analysis

Definition 4.1 (Read Disturb). Threshold voltage shift due to repeated read operations:

$$\Delta V_{thRD} = f(N_{reads}, V_{read}, V_{pass}, T, t_{cell}) \quad (16)$$

Electric field during read:

$$E_{read} = \frac{V_{pass} - V_{channel}}{8 - 10 \text{ nm}} \approx \frac{5 - 7 \text{ V}}{8 - 10 \text{ nm}} \sim 6 - 8 \text{ MV/cm} \quad (17)$$

• Electron Injection Mechanism

Low-level electron injection rate:

$$\frac{dQ}{dt} = A \cdot E^n \exp\left(-\frac{B}{E}\right) \cdot f(T) \quad (18)$$

Cumulative charge injection:

$$Q_{total} = \int_0^{N_{reads}} \frac{dQ}{dt} \cdot t_{read} dN \quad (19)$$

• Read Disturb Modeling

Arrhenius model:

$$\text{Rate}_{RD} = A_0 \exp\left(-\frac{E_a}{kT}\right) \cdot g(V_{pass}) \quad (20)$$

Power-law dependence:

$$N_{fail} = N_0 \left(\frac{V_{stress}}{V_{ref}}\right)^{-n} \quad (21)$$

5 4.5 Read Retry and Optimization

• Read Retry Algorithm

Theorem 5.1 (Adaptive Read Voltage). Optimal read voltage adjustment:

$$V_{read,opt} = V_{read,initial} + \Delta V \cdot \text{sign}(\text{Error pattern}) \quad (22)$$

Voltage step optimization:

$$\Delta V_{step} = \frac{\sigma_{distribution}}{k_{steps}} \quad \text{where } k_{steps} = 5 - 10 \quad (23)$$

• Threshold Tracking

Real-time threshold tracking:

$$\hat{V}_{th}(t) = V_{th}(t_0) + \int_{t_0}^t \frac{dV_{th}}{dt'} dt' \quad (24)$$

Drift rate estimation:

$$\frac{dV_{th}}{dt} = \alpha \log(t) + \beta \sqrt{t} + \gamma \quad (25)$$

- **Soft Information Generation**

Log-likelihood ratio (LLR):

$$\text{LLR} = \ln \left(\frac{P(\text{bit} = 0|Y)}{P(\text{bit} = 1|Y)} \right) \quad (26)$$

Soft decision based on multiple reads:

$$\text{Soft value} = \sum_{i=1}^{N_{reads}} w_i \cdot \text{Hard decision}_i \quad (27)$$

6 4.6 Advanced Sensing Techniques

- **All-Bit-Line Sensing**

Parallel sensing advantage:

$$\text{Throughput}_{ABL} = \frac{N_{BL} \cdot \text{Page size}}{t_{sense} + t_{transfer}} \quad (28)$$

Noise correlation in ABL:

$$\sigma_{noise,corr} = \sqrt{\sigma_{individual}^2 + \sigma_{common}^2 \cdot N_{BL}} \quad (29)$$

- **Multi-Level Sensing**

Simultaneous multi-level read:

$$\text{Data}[i] = \sum_{j=1}^{N_{levels}} 2^{j-1} \cdot \text{Compare}(I_{cell}, I_{ref,j}) \quad (30)$$

Sensing time for n -bit MLC:

$$t_{sense,MLC} = (2^n - 1) \cdot t_{sense,SLC} + t_{overhead} \quad (31)$$

7 4.7 Noise and Reliability

- **Noise Sources**

Total noise power:

$$\sigma_{total}^2 = \sigma_{thermal}^2 + \sigma_{shot}^2 + \sigma_{flicker}^2 + \sigma_{RTS}^2 \quad (32)$$

Thermal noise:

$$\sigma_{thermal}^2 = 4kTR_{on}B_{noise} \quad (33)$$

Shot noise:

$$\sigma_{shot}^2 = 2qI_{dc}B_{noise} \quad (34)$$

- **Signal-to-Noise Ratio**

SNR requirement:

$$\text{SNR} = \frac{|I_{on} - I_{off}|^2}{\sigma_{noise}^2} > 20 \text{ dB (minimum)} \quad (35)$$

Bit error rate estimation:

$$\text{BER} \approx \frac{1}{2} \text{erfc} \left(\frac{\text{Signal margin}}{2\sigma_{noise}} \right) \quad (36)$$

8 Examples

Example 8.1 (Sensing Time Calculation). Calculate minimum sensing time for: - Bit line capacitance: 2 pF - Cell current: 1 A (on), 10 nA (off) - Required voltage margin: 100 mV

Solution:

$$\Delta V_{sense} = \frac{I_{cell} \cdot t_{sense}}{C_{BL}} \quad (37)$$

$$100 \times 10^{-3} = \frac{1 \times 10^{-6} \cdot t_{sense}}{2 \times 10^{-12}} \quad (38)$$

$$t_{sense} = \frac{100 \times 10^{-3} \times 2 \times 10^{-12}}{1 \times 10^{-6}} \quad (39)$$

$$= 200 \text{ ns} \quad (40)$$

For off-state verification:

$$t_{sense,off} = \frac{100 \times 10^{-3} \times 2 \times 10^{-12}}{10 \times 10^{-9}} \quad (41)$$

$$= 20 \text{ s} \quad (42)$$

Example 8.2 (Read Disturb Analysis). Estimate read disturb for 10 reads: - Pass voltage: 5V, Temperature: 85°C - Activation energy: 1.1 eV - Time per read: 25 s

Solution:

$$E_{field} = \frac{5}{8 \times 10^{-9}} = 6.25 \times 10^8 \text{ V/m} \quad (43)$$

$$T = 85 + 273 = 358 \text{ K} \quad (44)$$

$$\text{Total time} = 10^6 \times 25 \times 10^{-6} = 25 \text{ s} \quad (45)$$

Using typical parameters $A = 10^{15}$, $B = 2.6 \times 10^8$:

$$\text{Rate} = 10^{15} \exp \left(-\frac{2.6 \times 10^8}{6.25 \times 10^8} \right) \quad (46)$$

$$\approx 10^{15} \exp(-0.42) \approx 6.6 \times 10^{14} \quad (47)$$

Negligible disturb expected for practical conditions.

9 Homework Problems

Assignment 4.1: Read Operations and Sensing

1. Sense Amplifier Design:

- Design for 1 A signal current
- Bit line capacitance: 3 pF
- Target sensing time: ≤ 1 ns
- Include offset cancellation

2. Reference Current Optimization:

- SLC with $I_{on}/I_{off} = 10^4$
- Process variation: $\pm 20\%$
- Temperature range: -40°C to 85°C
- Design tracking reference

3. Multi-Level Read:

- TLC read with 7 reference voltages
- Optimize reference spacing
- Calculate total read time
- Include bit line settling effects

4. Read Disturb Modeling:

- 10 read cycles specification
- Pass voltage optimization (4V-8V)
- Temperature acceleration modeling
- Design read voltage limits

5. Noise Analysis:

- Calculate thermal, shot, and $1/f$ noise
- Bandwidth: 10 MHz
- Load resistance: 10 k Ω
- Determine SNR requirements

6. Read Retry Algorithm:

- Implement hill-climbing algorithm
- 7 retry voltages, $\pm 0.5\text{V}$ range
- Minimize total read time
- Include ECC failure detection

7. Threshold Tracking:

- Model retention-induced drift
- Program/erase cycling effects
- Real-time tracking algorithm
- Predict optimal read voltages

8. **High-Speed Sensing:**

- Target: 1 GB/s read throughput
- Page size: 16 KB
- Design pipeline architecture
- Analyze power consumption

9. **3D NAND Reading:**

- String resistance effects
- Layer-to-layer variation
- Word line coupling analysis
- Compensation techniques

10. **Machine Learning Enhancement:**

- Neural network for threshold prediction
- Training data requirements
- Real-time inference complexity
- Performance improvement analysis

Next Chapter Preview: Chapter 5 will cover "Reliability and Wear Mechanisms" including endurance modeling, retention analysis, and statistical reliability frameworks.